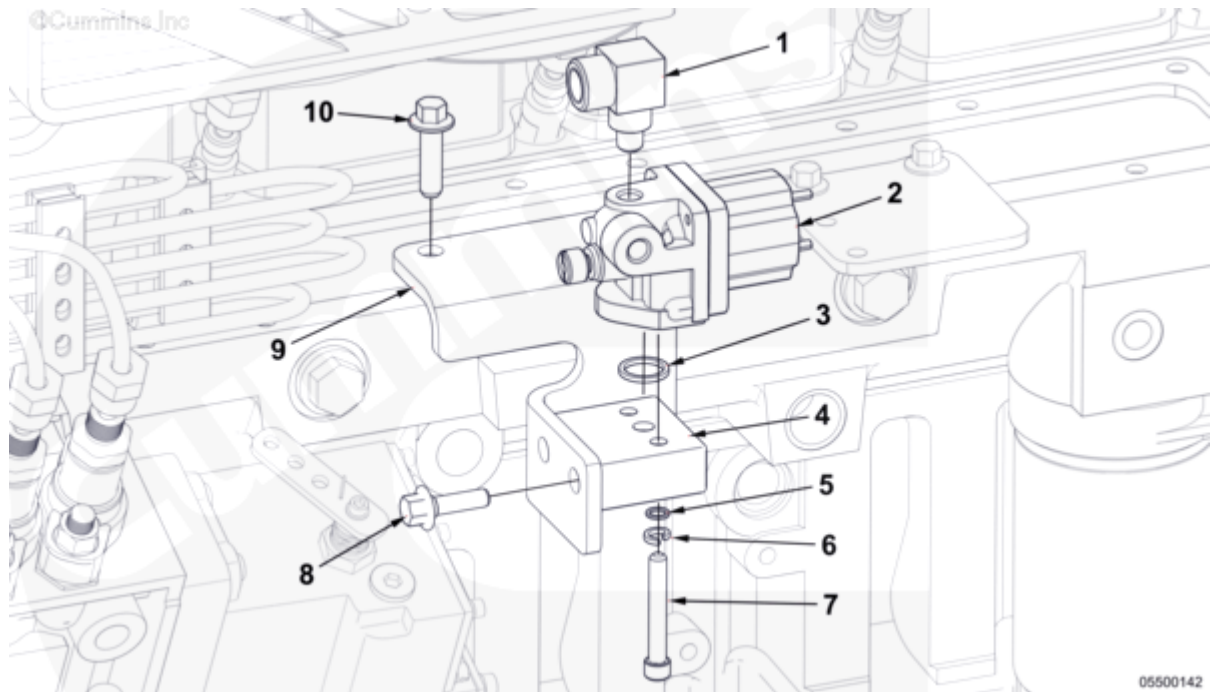


Trainings ▾

Exploded View



05500142

1. Fuel shutoff valve connector - 1
2. Fuel shutoff valve - 1
3. O-ring-1
4. Fuel shutoff valve adapter - 1
5. Plain washer - 2
6. Lock washer - 2
7. Mounting capscrew - 2
8. Fuel shutoff valve adapter capscrew - 2
9. Fuel shutoff valve bracket - 1
10. Fuel shutoff valve bracket capscrew - 2.

Select Service Tools

Recommended Cummins® Service Tools

- Volt-ohm Meter, Part Number 3164488
- Tamper Screw Removal Tool, Part Number 3399870.

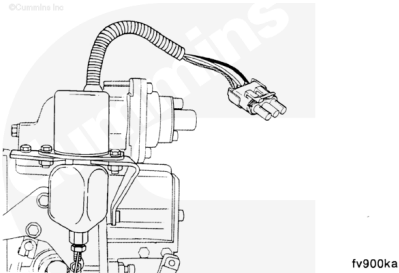
Additional Service Items

- No addition service items required.

General Information

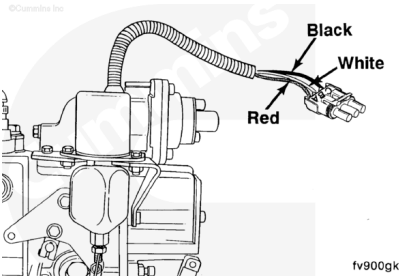
Shutdown Solenoid Troubleshooting (In-Line-Type Injection Pump)

- Engines using the in-line type of injection pumps are equipped with the Synchro-Start or Woodward™ fuel shutoff solenoid to actuate the shutoff lever. Both 12 VDC and 24 VDC external fuel shutoff solenoids are available.



Engines using the in-line type of injection pumps are equipped with the Synchro-Start or Woodward™ fuel shutoff solenoid to actuate the shutoff lever.

Both 12 VDC and 24 VDC external fuel shutoff solenoids are available.



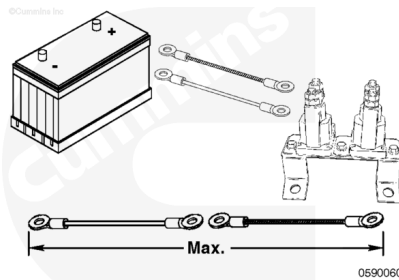
Color	Description	Weather-Pack™ Port
Black	Ground	C
White	Pull-in	B
Red	Hold-in	A

Use the chart below to find the correct gauge size and length of continuous wire for the white (pull-in) wire which connects to the solenoid wiring.

This is the total wire length from the battery to the solenoid and back to the battery. Both white and black wire length **must** be added.

Fourteen-gauge wire is required for the red (hold-in) wire, which connects to the “Run” terminal on the ignition switch.

The black (ground) wire **must** be the same size as the white (pull-in) wire.



Length of Wire - Maximum Length

Gauge	12 VDC	24 VDC
14	1.5 mm [5 ft]	2.7 mm [9 ft]
12	2.7 mm [9 ft]	4.3 mm [14 ft]
10	4.3 mm [14 ft]	7.0 mm [23 ft]

Preparatory Steps

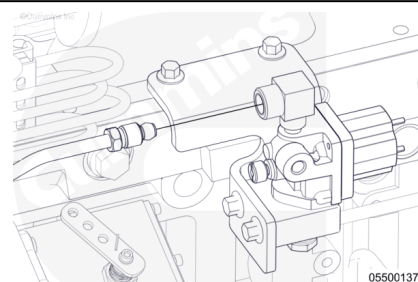
Bosch™ A Pump with RSV Governor

Note : This repair can be performed without removing the fuel pump from the engine.

- Removal of the shutoff lever, filter, and supply line is **not** necessary, if the solenoid can be accessed from the bottom of the fuel pump.
- Remove the fuel filter and fuel supply line, if necessary. Refer to Procedure 006-015 (</qs3/pubsys2/xml/en/procedures/41/41-006-015-tr.html>) and Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/41/41-006-024.html>)
- Label and remove the wiring harness from the shutoff solenoid.

Generator Set Applications

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the wire harness from the fuel shutoff valve.
- Remove the fuel supply line. Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/41/41-006-024.html>)



Initial Check

In-line Fuel Injection Pumps

⚠ WARNING ⚠

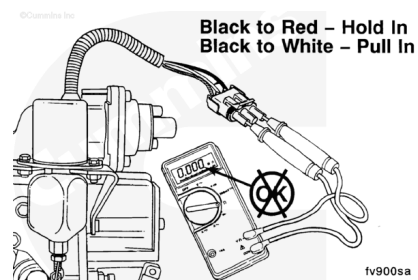
Wear protective clothing to reduce the possibility of personal injury. Solenoid surface temperature can exceed 175°C [347°F], which can cause serious burns to the skin in the

event of contact.

Note : The following check is for all In-line fuel injector pumps.

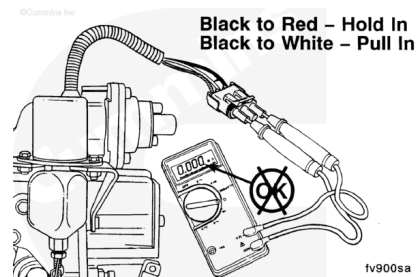
Values are taken at 20°C [68°F] and rated voltage. Minimum values are for 25-mm [1.00-in] maximum plunger travel. As the temperature of the solenoid increases, the voltage and resistance requirements increase, while the amperage requirements decrease.

The solenoid resistance can be checked using a multimeter. Disconnect the wiring harness and check the solenoid resistance.



Synchro-Start Solenoids 44.5 mm [1.75 in] Diameter Coil Canister

Solenoid Voltage	Acceptable Resistance Range in Ohms	
	Pull-in	Hold-in
12	0.198 to 0.242	10.00 to 12.21
24	0.738 to 0.902	37.17 to 45.43



Synchro-Start Solenoids 50.8 mm [2.00 in] Diameter Coil Canister

Solenoid Voltage	Acceptable Resistance Range in Ohms	
	Pull-in	Hold-in
12	0.175 to 0.213	12.75 to 15.56
24	0.554 to 0.678	46.76 to 57.15

Woodward™ solenoids with a 44.5 mm [1.75 in] Diameter Coil Canister

Solenoid Voltage	Acceptable Resistance Range in Ohms	
	Pull-in	Hold-in
12	0.232 to 0.284	9.818 to 12.001
24	0.878 to 1.073	46.548 to 56.892

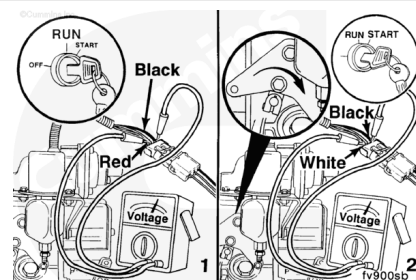
**Woodward™ solenoids with a 50.8 mm [2.00 in] Diameter
Coil Canister**

Solenoid Voltage	Acceptable Resistance Range in Ohms	
	Pull-in	Hold-in
12	0.175 to 0.213	12.75 to 15.59
24	0.554 to 0.678	46.76 to 57.15

Voltage Checking

Note : The following check is for all In-line fuel injector pumps.

To perform the solenoid voltage check, disconnect the solenoid rod end from the shutdown lever on the fuel pump, connect the wiring harness, and apply voltage to the solenoid with the ignition key as follows:



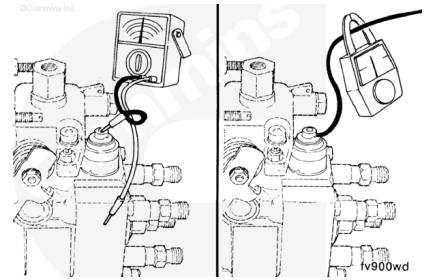
1. With the key in the RUN position, check the hold-in voltage.
2. With the shutdown lever held in the shutdown position, move the key to the START position, and check the pull-in voltage.

Use the table for Synchro-Start voltage specification with solenoid at 20°C [68°F]. Voltage requirements will be higher as engine temperature rises; therefore, these values are **only** valid with the solenoid at normal ambient temperatures.

Synchro-Start Solenoids 20° C [68°F]		
Solenoid Voltage	Minimum Voltage	
	Pull-in	Hold-in
12	8.5	5.2
24	17.0	9.4

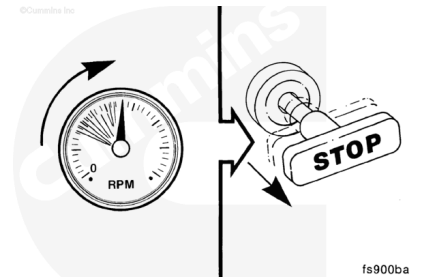
Bosch™ VE

Bosch™ Shutdown Values	Resistance Ohms	Peak Amperes
12 VDC	7.4 + 0.5	2
24 VDC	29.5 + 2.5	1



⚠ CAUTION ⚠

Do not connect the electrical wire to the solenoid when the plunger has been removed. Without the plunger, the valve can be damaged.

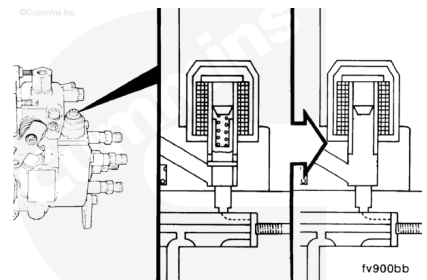


Malfunctioning valves and electrical wiring to the valve can be diagnosed by removing the plunger and spring, and then reinstalling the solenoid.

If the engine will start without the valve, the valve or the wiring to the valve is malfunctioning.

This method of removing the plunger to start the engine can be used, if necessary, to move the equipment to a service location.

Use the mechanical shutdown lever to stop the engine.



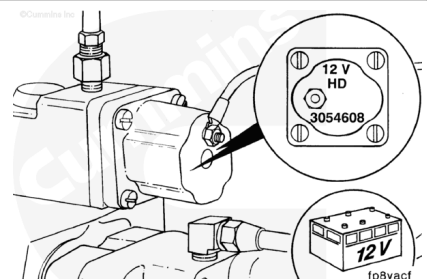
Generator Set Applications

Note : There must be only one wire connected to the shutoff valve.

Connect the wire harness to the shutoff valve coil.

Make sure the voltage is correct.

The coil required voltage and part number are cast into the terminal connection case at the end of the shutoff valve.

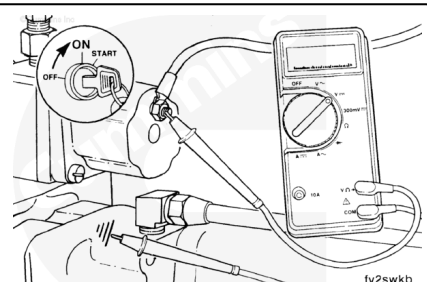


Turn the key switch to the "ON" position.

Check the DC voltage of the coil with a volt-ohm meter, Part Number 3164488, or equivalent.

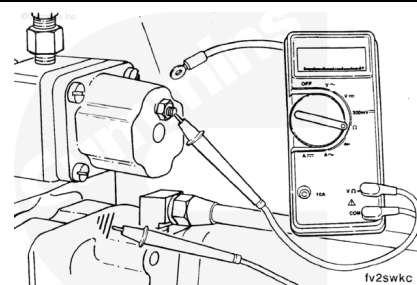
The voltage **must** be same as the voltage specifications.

Turn the key switch to the "OFF" position.



Make sure the coil wire is **not** connected before checking the shutoff valve resistance.

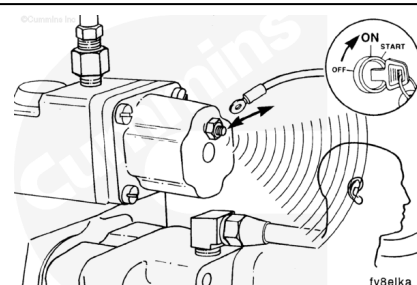
Check the resistance with a volt-ohm meter. The resistance **must** be:



Coil Resistance	
Voltage	Resistance
24 VDC	28 to 32 ohms

Replace the shutoff valve if the resistance does **not** meet the specifications.

Turn the key switch to the "ON" position. Listen for the valve to "click" when the wire is touched to the coil terminal. If the valve does **not** "click", repair or replace the fuel shutoff valve.

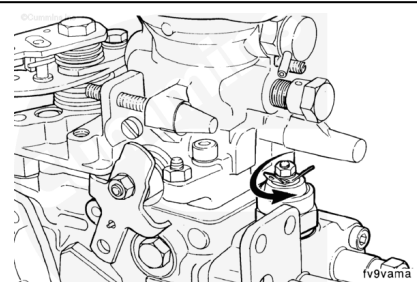


Remove

Bosch™ VE

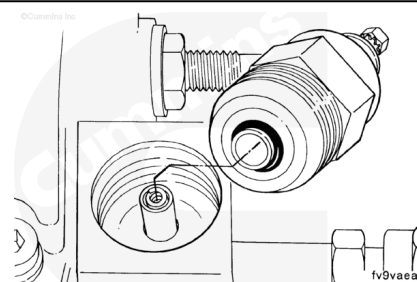
Remove the valve.

Note : The Bosch™ VE valve is shown. The valve for Lucas™ CAV is located at the bottom of the pump.



⚠ CAUTION ⚠

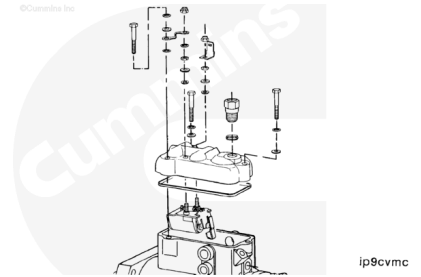
When removing the valve, be careful not to drop the plunger and spring. Doing so can result in fuel pump damage.



Stanadyne DB4

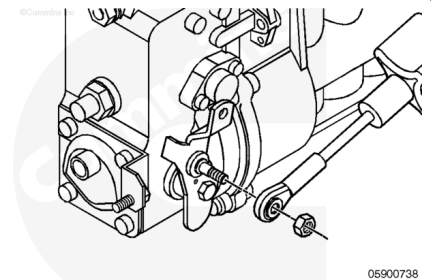
Tamper Screw Removal Tool, Part Number 3399870.

- Remove the electrical wiring.
- Remove the fuel drain line. Refer to Procedure 006- 021 in Section 6.
- Remove the throttle and shutoff linkage. See equipment manufacturer service information.
- Remove tamper-resistant screws using service tool kit, Part Number 3399870. Refer to Procedure 005-012 in Section 5. (/qs3/pubsys2/xml/en/procedures/41/41-005-012.html)
- Remove the fuel injection pump top cover. Reference the Master Repair Manual, Injector Pumps and Injectors, Bulletin 3666037.

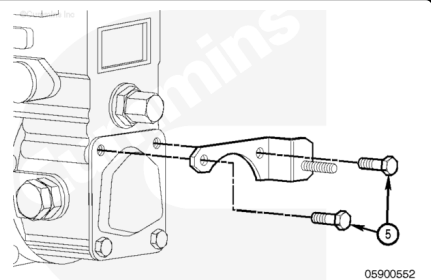


Bosch™ A Pump with RSV Governor

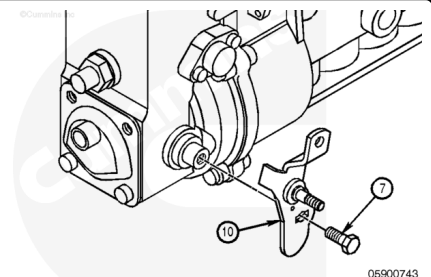
Remove the rod end from the shutoff lever.



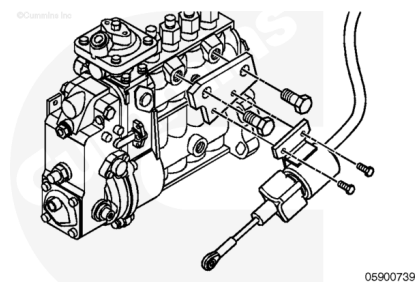
Remove the stop screw and bracket assembly.



Remove the shutoff lever (10) over the shutoff shaft on the inboard side of the fuel pump.



Remove the solenoid from the solenoid bracket assembly and solenoid bracket assembly from the fuel pump. If the old 1-3/4 inch solenoid and bracket assembly is being replaced with the new 2-inch solenoid, discard the solenoid mounting capscrews.



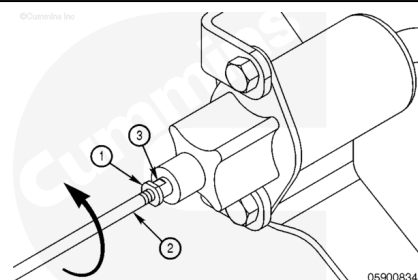
Bosch™ P Pump with RQVK Governor

The solenoid does **not** have to be removed from engine to replace the control rod.

Loosen the locknut washer (1) from the solenoid (3).

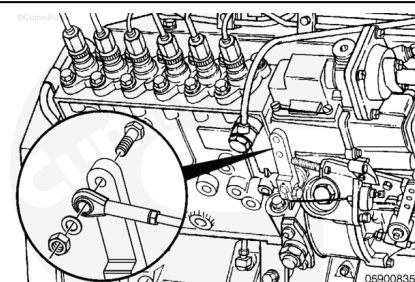
Disconnect the control rod (2) at the lever if control rod is **not** broken.

Unscrew the control rod while holding the solenoid swivel.



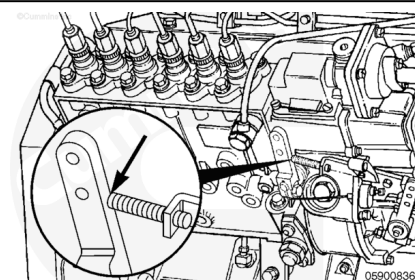
Loosen the capscrew and nut that holds the rod end onto the shutoff lever.

Retain the spacer between the rod end and the shutoff lever.



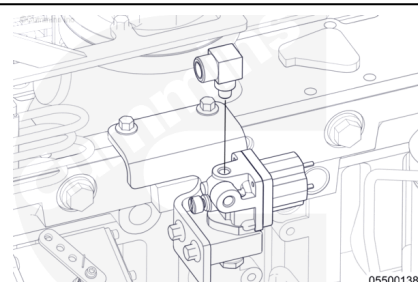
Check the shutoff lever stop lever stop bracket.

The lever **must** make contact with the stop screw on the stop screw bracket. If the lever does **not** touch the stop screw, adjust the stop screw out 1-1/2 turns past the point of contact between the stop screw and the shutoff lever.

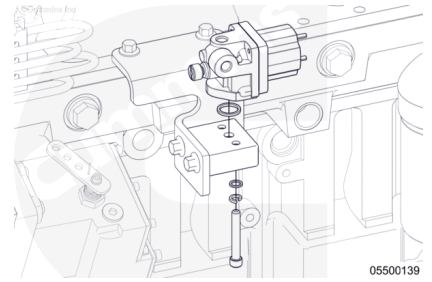


Generator Set Applications

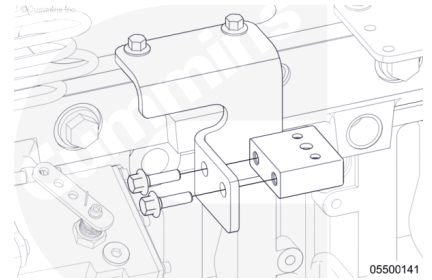
Remove the fuel shutoff valve connector.



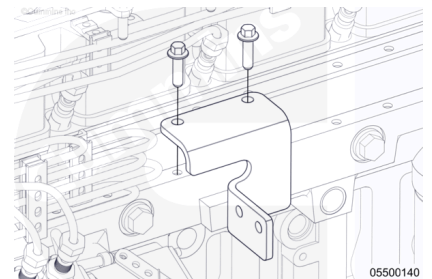
Remove the fuel shutoff valve, o-ring, lock washer, plain washer and capscrews.



Remove the fuel shutoff valve adapter.

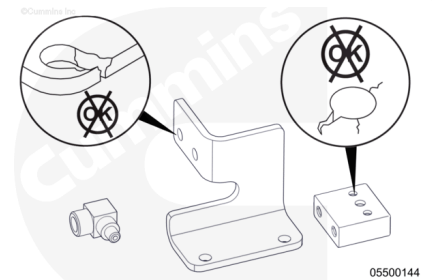


Remove the fuel shutoff valve bracket and capscrews.



Clean and Inspect for Reuse

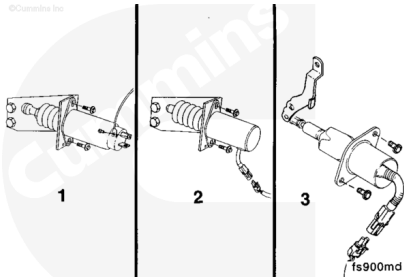
Replace the fuel shutoff valve bracket, connector and adapter, if any crack or damage found.



Install

Note : Make sure the acorn nut is tightened to be snug on the fuel shutoff solenoid shaft (Synchro-start only).

Install the new fuel shutoff solenoid to the bracket, and connect the wires. Make sure the wiring harness one the Trombetta solenoid is installed in the six-o'clock position.

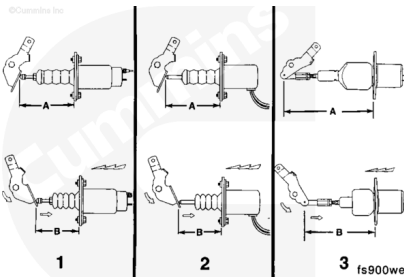


Torque Value: 10 n•m [89 in-lb]

- 1. Synchro-start
- 2. Trombetta
- 3. Direct link.

Activate the switch and check the plunger travel.

	Synchro-start	Trombetta	Direct link
A =	86.8.2 mm [3.4 in]	91.4 mm [3.6 in]	
B =	60.2 mm [2.4 in]	63.5 mm [2.5 in]	117.1 mm [4.61 in]

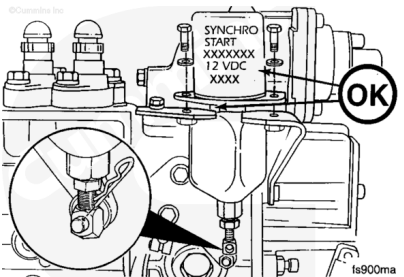


The plunger **must** be retracted when the fuel shutoff solenoid is activated to the RUN position B. The fuel shutoff solenoid **must** operate without binding.

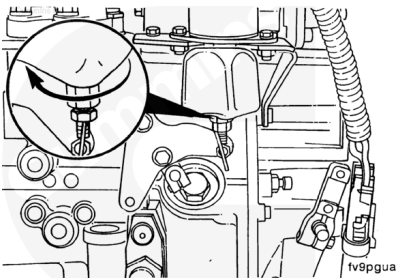
Remove the hitch pin clip, the mounting capscrews, and the fuel shutoff solenoid.

Install the new solenoid in reverse order of removal, and connect the wires.

Torque Value: 10 n•m [89 in-lb]

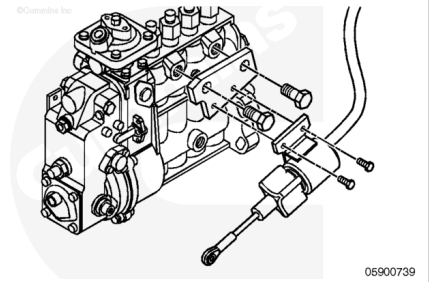


Adjust the solenoid linkage as necessary so that the plunger is magnetically held in with the shutoff lever in the absolute full-run position. Turn the large hex nut on the end of the plunger to make adjustments, and secure in place with a locknut.



⚠ CAUTION ⚠

Failure to observe proper cranking and fuel system priming procedures can cause solenoid malfunction.



⚠ CAUTION ⚠

New solenoid mounting capscrews must be used if replacing an old 1-3/4-inch solenoid with a 2-inch solenoid. Insufficient thread engagement can cause damage to the pump and the solenoid.

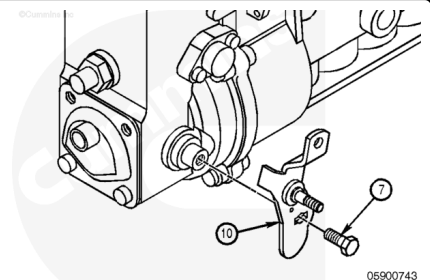
Install the shutoff solenoid on the fuel pump using two new M16 x 1.5-16 capscrews. Apply Loctite® to the capscrew threads. Tighten the capscrews just enough to hold the solenoid in place.

Note : New solenoid mounting capscrews have threadlocker pre-applied. Loctite® application is not necessary when new capscrews are installed.

Install the shutoff lever (10) over the shutoff shaft on the fuel pump.

Use the capscrew (7) previously removed to hold the shutoff lever in place.

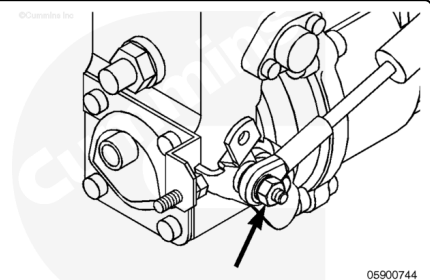
Torque Value: 9 n•m [80 in-lb]



Connect the rod end of the solenoid to the shutoff lever.

Tighten the shutoff lever nut.

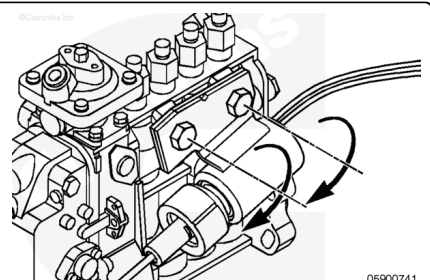
Torque Value: 9 n•m [80 in-lb]



⚠ CAUTION ⚠

The solenoid mounting capscrews must not be overtightened. Distortion to the fuel pump body can result causing the rack to stick in the fuel pump.

Tighten the solenoid mounting capscrews.

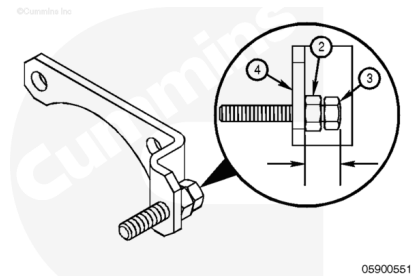


Torque Value: 24 n·m [212 in-lb]

Note : If the stop screw is **not** moved, readjustment may **not** be necessary.

Assemble the stop screw to the stop screw bracket. Adjust the stop screw (3) so the contact surface of the screw protrudes 10 mm [0.394 in] beyond the surface of the stop bracket (4).

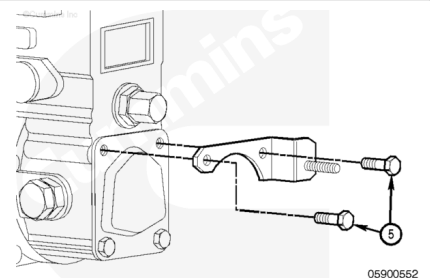
Tighten the nut (2) against the stop bracket to lock the stop screw in place.



Install the stop bracket assembly using two M6 x 1-16 capscrews (5).

Tighten the capscrews.

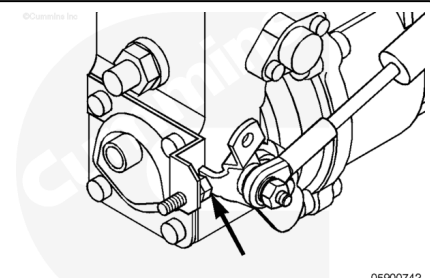
Torque Value: 7 n·m [62 in-lb]



⚠ CAUTION ⚠

The solenoid rod length is pre-set. Adjustment of the solenoid rod can cause damage to the solenoid or lead to a low power complaint.

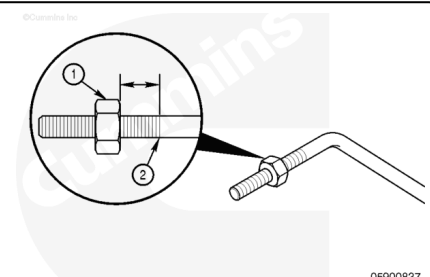
Verify the shutoff lever is contacting the stop screw and is in the full stop position. If the shutoff lever does **not** contact the stop screw, verify the shutoff solenoid and stop screw bracket are assembled correctly. Loosen the solenoid mounting capscrews and readjust the solenoid, if necessary.



Bosch™ P Pump with RQVK Governor

The rod end to control rod orientation is important. If the rod end is installed incorrectly, improper operation of the shutoff solenoid assembly can occur.

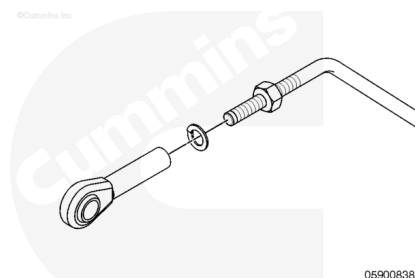
Install the locknut (1) onto the new control rod at the rod end until there is 13 mm [7/16 inch] between the start of the threads (2) and nut (1).



Install the lock washer.

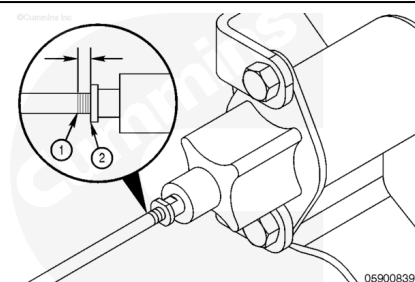
Screw the rod end onto the control rod, hand-tighten and make certain **not** to move the locknut. Lay the control rod and rod end on a flat surface. Rotate the rod end so both the control rod and the rod end lay flat.

Tighten the locknut onto the rod end.



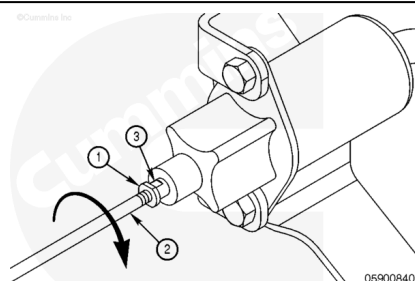
Install the locknut (2) on the control rod at the solenoid end until there is 6 mm [1/4 inch] between start of the threads (1) and locknut (2).

Install the lock washer and screw the control rod into the solenoid.



Install the control rod end capscrew onto the fuel pump shutoff lever to align the control rod.

Tighten the locknut (1) while holding the solenoid swivel (3).

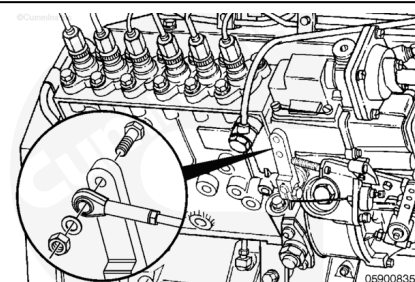


The solenoid is designed to allow rotation of the control rod.

Make certain to install the spacer between the rod end and the shutoff lever.

Tighten the capscrew and nut that holds the rod end onto the shutoff lever.

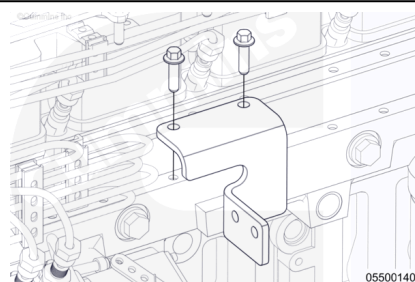
Torque Value: 9 n•m [80 in-lb]



Generator Set Applications

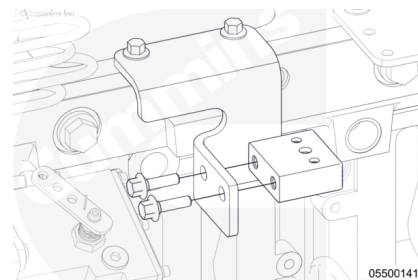
Install the fuel shutoff valve bracket.

Torque Value: 43 n•m [32 ft-lb]

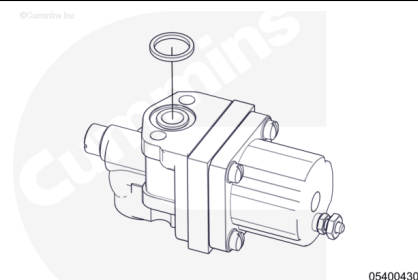


Install the fuel shutoff valve adapter.

Torque Value: 43 n•m [32 ft-lb]



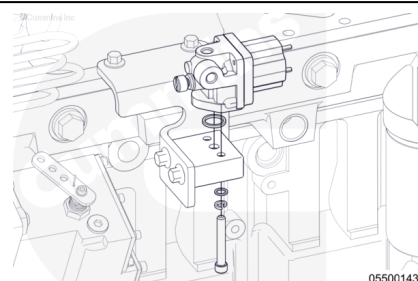
Install the O-ring to the groove.



Install the fuel shutoff valve, lock washer, plain washer.

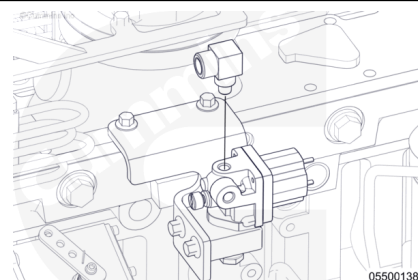
Tighten the capscrews.

Torque Value: 13 n•m [115 in-lb]



Install the fuel shutoff valve connector.

Torque Value: 6 n•m [53 in-lb]



Finishing Steps

- Install all components removed, including the fuel filter and fuel supply line, if removed. Refer to Procedure 006-015 (</qs3/pubsys2/xml/en/procedures/41/41-006-015-tr.html>) and Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/41/41-006-024.html>)
- Connect the wire harness to the fuel shutoff solenoid.
- Start the engine and check for proper operation.

